AI501 Part I Exam June 2025

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3 point

Which type of environment are most suitable for AND–OR trees?

- ○ Unknown environments
- ○ Nondeterministic environments
- ○ Online environments
- ○ Non-observable environments

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3 point

What is a key characteristic of 'general AI' as discussed in Chapter 28 Philosophy, Ethics, and Safety of AI?

- ○ Limited to predefined functions
- ○ Capable of performing any intellectual task that a human can do
- ○ Dependent on specific programming
- ○ Specialized in narrow tasks

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3 point

Which algorithm extends the minimax approach to handle games involving elements of chance?

- Expectiminimax algorithm
- A*
- Monte Carlo Tree Search
- Alpha-beta pruning

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3 point

What does the soundness property ensure in propositional logic?

- If a sentence can be derived using inference rules, then it must be true in all models.
- Any two logically equivalent sentences must have the same truth value.
- Every logical system has at least one model.
- If a sentence is true in all models, then it can be derived using inference rules.

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3 point

In First-Order Logic, what does the expression $\forall x\,(P(x) \vee Q(x))$ mean precisely?

- For all x, $P(x)$ and $Q(x)$ are both true.
- There exists an x for which either $P(x)$ or $Q(x)$ is true.
- For all x, either $P(x)$ is true or $Q(x)$ is true (or both).
- There exists an x for which $P(x)$ and $Q(x)$ are both true.

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3 point

For a self-driving car, which of the following is considered a sensor?

- Speedometer.
- Steering wheel.
- Engine.
- Brake pedal.

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3 point

What is the primary difference between informed and uninformed search strategies?

- ○ Uninformed strategies guarantee optimal solutions.
- ○ Uninformed strategies require less memory.
- ○ Informed strategies use heuristics to guide the search.
- ○ Informed strategies explore all possible nodes.

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3 point

For which type of games is the minimax algorithm particularly well-suited?

- ○ Two-player, zero-sum, turn-based games with perfect information
- ○ Single-player puzzle games
- ○ Real-time strategy games
- ○ Multiplayer games with hidden information

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3 point

What does the existential quantifier (*exists*) signify in First-Order Logic?

- ○ That a statement is true for all possible instances.
- ○ That a statement is true and false simultaneously.
- ○ That there exists at least one instance for which the statement is true.
- ○ That a statement is false for all instances.

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3 point

Which of the following is a key feature of the Planning Domain Definition Language (PDDL)?

- ○ It is limited to representing only stochastic planning problems.
- ○ It is designed to standardize the representation of planning problems across different planners.
- ○ It does not support the definition of action preconditions and effects.
- ○ It is primarily used for representing neural network configurations.

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3 point

Which of the following statements is true?

- Every contradiction is satisfiable.
- No tautology is satisfiable.
- Every satisfiable statement is a tautology.
- Every tautology is satisfiable.

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3 point

Which of the following represents the logical AND operation?

- $\wedge$
- $\neg$
- $\vee$
- $\rightarrow$

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3 point

What is the primary goal of the Turing Test in the context of AI?

- ○ To evaluate a machine's ability to exhibit intelligent behavior indistinguishable from a human
- ○ To assess a machine's ability to solve mathematical problems
- ○ To measure a machine's processing speed
- ○ To determine a machine's capacity for learning

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3 point

Which of the following environments is both deterministic and fully observable?

- ○ Self-driving car navigation
- ○ Stock market
- ○ Poker game
- ○ Chess game

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3 point

The "Disability Argument" in artificial intelligence fundamentally challenges the notion that:

- AI systems will inevitably surpass human intelligence.
- Machine learning can improve without explicit programming.
- AI can fully replicate uniquely human traits such as creativity and emotions.
- Machines can simulate human cognitive processes effectively.

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3 point

Which of the following best describes the sum rule (addition rule) in probability theory?

- It calculates the probability of the intersection of two events.
- It determines the probability of the union of two events.
- It finds the probability of two independent events occurring together.
- It computes the probability of the complement of an event.

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3 point

Which of the following environments is considered entirely episodic?

- Teaching a class
- Stock market trading
- Image classification tasks
- Autonomous driving

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3 point

What would be an appropriate performance measure for an AI designed to play a strategy game?

- The speed at which the AI makes decisions.
- The win/loss ratio over multiple games.
- The complexity of the algorithms used.
- The graphical quality of the game interface.

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3 point

Which of the following is a limitation of the standard minimax algorithm?

- It is inefficient for games with large search trees due to exhaustive exploration.
- It is only applicable to games with imperfect information.
- All of the other answers
- It can not easily be adapted to handle games with more than two players.

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3 point

What is the difference between syntax and semantics?

- Semantics focuses on sentence formation rules, and syntax determines truth values.
- Syntax defines the structure and formation rules of sentences, while semantics assigns meaning by defining the truth of sentences in a given world.
- Syntax and semantics both define the meaning of logical connectives.
- Syntax determines the truth of a sentence in a world, while semantics describes how sentences are formed.

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3 point

In a Constraint Satisfaction Problem (CSP), what does a 'variable' represent?

- A constraint that must be satisfied
- A function defining relationships between elements
- A possible value that can be assigned
- An element that can take on different values

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3 point

Which of the following formulas represents the sum rule for two events, A and B?

- $P(A \cap B) = P(A) \times P(B \mid A)$
- $P(A \cup B) = P(A) \times P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cap B) = P(A) + P(B)$

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3 point

What does the 'least-constraining value' heuristic aim to achieve in CSPs?

- Selects the value that rules out the most choices for neighboring variables
- Selects the value that rules out the fewest choices for neighboring variables
- Selects the value that is least frequently used
- Selects the value that is most frequently used

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3 point

What characterizes a stochastic environment?

- The environment has elements of randomness and unpredictability
- The environment does not change over time
- The next state of the environment is determined by the current state and the agent's action
- The environment is completely predictable

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3 point

Which of the following is NOT a characteristic of a rational agent?

- Makes decisions based on percept history
- Acts to achieve the best expected outcome
- Operates autonomously
- Always mimics human behavior

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3 point

What distinguishes 'classical planning' from other forms of planning in AI?

- It assumes a fully observable, deterministic, finite, static, and discrete environment.
- It incorporates learning mechanisms to adapt plans over time.
- It focuses on real-time dynamic planning scenarios.
- It deals exclusively with stochastic and partially observable environments.

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3 point

In the PEAS description of an agent, what does 'Actuators' refer to?

- They process sensory information.
- They execute actions that allow the agent to interact with the environment.
- They measure the performance of the agent.
- They define the environment for the agent.

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3 point

What is the main advantage of using 'heuristic search' in planning algorithms?

- It eliminates the need for defining action preconditions and effects.
- It guarantees finding the optimal plan in all scenarios.
- It possibly reduces the search space by providing estimates of the cost to reach the goal, leading to more efficient planning.
- It allows for planning in non-deterministic environments without any modifications.

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3 point

How would you represent the statement "There exists a person who is a student and is studying Computer Science" in First-Order Logic?

- ○ $\forall x\,(\text{Person}(x) \land \text{Student}(x) \land \text{Studies}(x, \text{ComputerScience}))$
- ○ $\exists x\,(\text{Person}(x) \lor \text{Student}(x) \lor \text{Studies}(x, \text{ComputerScience}))$
- ○ $\forall x\,(\text{Person}(x) \rightarrow \text{Student}(x) \rightarrow \text{Studies}(x, \text{ComputerScience}))$
- ○ $\exists x\,(\text{Person}(x) \land \text{Student}(x) \land \text{Studies}(x, \text{ComputerScience}))$

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3 point

In hierarchical planning, what is the purpose of 'task decomposition'?

- ○ To eliminate unnecessary tasks from the plan.
- ○ To sequence tasks in chronological order.
- ○ To assign tasks to different agents in a multi-agent system.
- ○ To break down complex tasks into simpler, more manageable subtasks.